Supplementary Materials

CRVE: Low-Risk, LLM-Free Query Expansion for Lexical Retrieval from Stored Corpus Contexts

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Section S1 (Tables S1–S4) and Section S2 (Tables S5 and S6) accompany the paper. Section, table and figure numbers without the prefix S refer to the paper. Elsewhere, S1 and S2 name the chunk-selection scores of the context sidecar (Section 3.2 of the paper): "CRVE S1" and "CRVE S2" are CRVE with the sidecar chunks chosen by that score. Code, stored results and the scripts that regenerate every table are in the repository at `https://github.com/spchjken/CRVE` (`paper/README.md` maps each table to its script and output file).

S1 Tuning on Both Sides

The baselines of Section 5 of the paper run at one fixed setting each. This comparison was declared before the chunk-selection score S2 was adopted; the readings for CRVE with S2 apply its family and reading rules post hoc. Here RM3 and Bo1 are tuned per collection on the evaluation queries themselves (oracle), which favours them, and by two-fold cross-validation; CRVE is tuned the same way over K and w. For RM3, tuning varies the number of feedback documents, the number of expansion terms and the interpolation weight λ. The grids differ in size: 12 settings for CRVE, 18 for RM3 and 6 for Bo1, and the oracles often chose settings at the edge of their grid ($K = 10$ and $w = 0.30$ for CRVE, $\lambda = 0.7$ for RM3). The query sets are capped at 500 queries per collection. Table S1 gives the comparison with tuned PRF summarised in Section 5.1 of the paper. When both sides are tuned on the test queries (oracle against oracle), CRVE is better or non-inferior in 14 of 16 contrasts and never worse (Table S2); with S2 alone, in 7 of 8. In each of the four settings, cross-validated tuning adds less on average to CRVE with S2 than to RM3 and to Bo1 (Table S3), although against Bo1 the ranges over the 30 splits overlap in three of the four settings (both over DPH and over BM25 over 1M): averaged over 30 random splits, it adds between -0.0001 and $+0.0036$ nDCG@10 to the fixed setting of CRVE with S2, against $+0.0024$ to $+0.0158$ for RM3 and Bo1. Tuned PRF still lowers several collections (Table S4). Per collection, CRVE (S2) at its fixed setting is ahead of the best-of-both oracle by more than 0.005 on 10 collections, behind it on 9 and within ± 0.005 on 6 over BM25, and 9, 10 and 6 over DPH.

Table S1: Difference in nDCG@10 between CRVE and RM3 or Bo1 tuned against it: oracle (parameters chosen on the test queries), cross-validated (CV) and best of both. Best of both: the better of the two oracles. CV (42): seed-42 split; CV (30): 30 splits. CRVE with the S2 chunks (default $K = 5$, $w = 0.20$); the comparison was declared before S2 was adopted, and these readings apply its family and reading rules post hoc; capped query sets (at most 500 queries per collection). Level: Holm-adjusted over four contrasts per group, 95%, or the range (smallest, largest) over 30 cross-validation splits. Rel.: percentage of the tuned PRF's nDCG@10.

CRVE minus	Difference	Interval	Level	Reading	Rel. (%)
Under 1M, base BM25					
RM3 oracle	+0.0100	[+0.0032, +0.0170]	98.75% Holm	better	+4.1
Bo1 oracle	−0.0019	[−0.0087, +0.0049]	97.5% Holm	unclear	−0.8
Best of both	−0.0074	[−0.0132, −0.0015]	95%	below zero	−2.8
RM3 CV (42)	+0.0127	[+0.0074, +0.0182]	95%	above zero	+5.3
Bo1 CV (42)	+0.0057	[−0.0004, +0.0118]	95%	uncertain	+2.3
RM3 CV (30)	+0.0142	[+0.0126, +0.0167]	range	above zero in every split	+6.0
Bo1 CV (30)	+0.0036	[−0.0001, +0.0073]	range	uncertain	+1.5
Under 1M, base DPH					
RM3 oracle	+0.0012	[−0.0046, +0.0069]	95% Holm	unclear	+0.4
Bo1 oracle	+0.0141	[+0.0070, +0.0217]	98.33% Holm	better	+5.4
Best of both	0.0000	[−0.0056, +0.0055]	95%	uncertain	0.0
RM3 CV (42)	+0.0065	[+0.0008, +0.0121]	95%	above zero	+2.4
Bo1 CV (42)	+0.0209	[+0.0143, +0.0275]	95%	above zero	+8.2
RM3 CV (30)	+0.0063	[+0.0047, +0.0082]	range	above zero in every split	+2.3
Bo1 CV (30)	+0.0217	[+0.0182, +0.0246]	range	above zero in every split	+8.5
Over 1M, base BM25					
RM3 oracle	+0.0039	[−0.0012, +0.0091]	97.5% Holm	non-inferior	+1.1
Bo1 oracle	+0.0104	[+0.0032, +0.0176]	98.33% Holm	better	+2.9
Best of both	+0.0027	[−0.0018, +0.0072]	95%	uncertain	+0.8
RM3 CV (42)	+0.0051	[+0.0006, +0.0095]	95%	above zero	+1.4
Bo1 CV (42)	+0.0116	[+0.0057, +0.0176]	95%	above zero	+3.3
RM3 CV (30)	+0.0051	[+0.0041, +0.0064]	range	above zero in every split	+1.4
Bo1 CV (30)	+0.0138	[+0.0117, +0.0156]	range	above zero in every split	+3.9
Over 1M, base DPH					
RM3 oracle	−0.0027	[−0.0078, +0.0023]	95% Holm	unclear	−0.7
Bo1 oracle	+0.0142	[+0.0060, +0.0226]	98.75% Holm	better	+3.6
Best of both	−0.0037	[−0.0087, +0.0014]	95%	uncertain	−0.9
RM3 CV (42)	+0.0003	[−0.0050, +0.0053]	95%	uncertain	+0.1
Bo1 CV (42)	+0.0157	[+0.0093, +0.0221]	95%	above zero	+4.0
RM3 CV (30)	−0.0009	[−0.0024, +0.0011]	range	uncertain	−0.2
Bo1 CV (30)	+0.0172	[+0.0142, +0.0233]	range	above zero in every split	+4.4

Holm rows keep the declared reading (better, non-inferior with lower bound above -0.002, or unclear); 95% rows and the ranges over the 30 splits read above zero, below zero or uncertain. A positive difference favours CRVE.

Table S2: Both sides tuned: difference in nDCG@10 between tuned CRVE (S1 or S2) and tuned RM3, tuned Bo1 or the best-of-both oracle. Δ: absolute difference; Both: best-of-both oracle. Oracle: parameters chosen on the test queries; CV: parameters chosen by cross-validation, with the seed-42 split (95% interval) and the mean of 30 splits with their smallest and largest value. Capped query sets (at most 500 queries per collection).

Sidecar / PRF	Oracle against oracle					CV against CV (seed 42)				CV, 30 splits	
	Δ	Interval	Level	Reading	(%)	Δ	Interval	Reading	(%)	Mean	Min, max
Under 1M, base BM25											
S1 / RM3	+0.0176	[+0.0098, +0.0257]	99.38% Holm	better	+7.3	+0.0132	[+0.0078, +0.0186]	above zero	+5.5	+0.0182	[+0.0157, +0.0209]
S1 / Bo1	+0.0057	[−0.0012, +0.0126]	97.5% Holm	non-inferior	+2.3	+0.0062	[+0.0000, +0.0124]	above zero	+2.5	+0.0076	[+0.0031, +0.0127]
S1 / both	+0.0003	[−0.0056, +0.0061]	95%	uncertain	+0.1						
S2 / RM3	+0.0149	[+0.0076, +0.0224]	99.29% Holm	better	+6.2	+0.0131	[+0.0077, +0.0186]	above zero	+5.5	+0.0149	[+0.0128, +0.0171]
S2 / Bo1	+0.0030	[−0.0028, +0.0090]	95% Holm	unclear	+1.2	+0.0061	[−0.0001, +0.0123]	uncertain	+2.5	+0.0043	[−0.0004, +0.0089]
S2 / both	−0.0025	[−0.0082, +0.0033]	95%	uncertain	−1.0						
Under 1M, base DPH											
S1 / RM3	+0.0081	[+0.0007, +0.0157]	98.75% Holm	better	+2.9	+0.0085	[+0.0025, +0.0143]	above zero	+3.1	+0.0092	[+0.0062, +0.0112]
S1 / Bo1	+0.0210	[+0.0131, +0.0294]	99.17% Holm	better	+8.0	+0.0229	[+0.0162, +0.0299]	above zero	+9.0	+0.0246	[+0.0208, +0.0278]
S1 / both	+0.0069	[+0.0014, +0.0125]	95%	above zero	+2.5						
S2 / RM3	+0.0070	[−0.0011, +0.0149]	98.33% Holm	non-inferior	+2.6	+0.0079	[+0.0018, +0.0140]	above zero	+2.9	+0.0076	[+0.0053, +0.0093]
S2 / Bo1	+0.0199	[+0.0117, +0.0286]	99% Holm	better	+7.6	+0.0223	[+0.0157, +0.0291]	above zero	+8.7	+0.0230	[+0.0194, +0.0263]
S2 / both	+0.0059	[−0.0003, +0.0120]	95%	uncertain	+2.1						
Over 1M, base BM25											
S1 / RM3	+0.0086	[+0.0028, +0.0145]	98.75% Holm	better	+2.4	+0.0078	[+0.0031, +0.0126]	above zero	+2.2	+0.0080	[+0.0055, +0.0098]
S1 / Bo1	+0.0151	[+0.0068, +0.0233]	99.38% Holm	better	+4.2	+0.0144	[+0.0084, +0.0205]	above zero	+4.1	+0.0167	[+0.0132, +0.0190]
S1 / both	+0.0074	[+0.0027, +0.0120]	95%	above zero	+2.0						
S2 / RM3	+0.0070	[+0.0017, +0.0125]	98.33% Holm	better	+1.9	+0.0042	[−0.0002, +0.0086]	uncertain	+1.2	+0.0050	[+0.0024, +0.0074]
S2 / Bo1	+0.0135	[+0.0056, +0.0215]	99.29% Holm	better	+3.8	+0.0108	[+0.0049, +0.0168]	above zero	+3.0	+0.0137	[+0.0100, +0.0162]
S2 / both	+0.0058	[+0.0014, +0.0103]	95%	above zero	+1.6						
Over 1M, base DPH											
S1 / RM3	+0.0018	[−0.0026, +0.0061]	95% Holm	unclear	+0.4	+0.0015	[−0.0029, +0.0059]	uncertain	+0.4	+0.0011	[−0.0014, +0.0045]
S1 / Bo1	+0.0187	[+0.0106, +0.0271]	99.17% Holm	better	+4.7	+0.0169	[+0.0110, +0.0231]	above zero	+4.3	+0.0192	[+0.0148, +0.0250]
S1 / both	+0.0008	[−0.0034, +0.0051]	95%	uncertain	+0.2						
S2 / RM3	+0.0035	[−0.0018, +0.0086]	97.5% Holm	non-inferior	+0.8	+0.0045	[−0.0000, +0.0089]	uncertain	+1.1	+0.0027	[+0.0001, +0.0050]
S2 / Bo1	+0.0204	[+0.0123, +0.0287]	99% Holm	better	+5.2	+0.0199	[+0.0140, +0.0260]	above zero	+5.1	+0.0208	[+0.0165, +0.0261]
S2 / both	+0.0025	[−0.0018, +0.0069]	95%	uncertain	+0.6						

Level: Holm-adjusted over eight contrasts per group, or 95% for the best-of-both oracle. Holm rows keep the declared reading (better, non-inferior, unclear); the other rows read above zero, below zero or uncertain. A positive difference favours CRVE. Percentage columns: difference as a percentage of the PRF's nDCG@10. Bounds printed as ±0.0000 are read from the unrounded values.

Table S3: Headroom from tuning: gain in nDCG@10 of the oracle and of cross-validated (CV) settings over the default settings, for CRVE (S1, S2), RM3 and Bo1. Default: nDCG@10 of the default setting; 95% query-bootstrap intervals for the oracle and the seed-42 split; for the 30 splits, the mean with the smallest and largest value; Rel.: percentage of the default level. Capped query sets (at most 500 queries per collection).

Method	Default	Oracle minus default			CV minus default (seed 42)			CV minus default, 30 splits		
	nDCG@10	Difference	Interval	Rel. (%)	Difference	Interval	Rel. (%)	Mean	Min, max	Rel. (%)
Under 1M, base BM25										
CRVE S1	0.251	+0.0077	[+0.0050, +0.0105]	+3.0	+0.0005	[−0.0019, +0.0030]	+0.2	+0.0040	[+0.0011, +0.0059]	+1.6
CRVE S2	0.251	+0.0049	[+0.0031, +0.0069]	+1.9	+0.0004	[−0.0015, +0.0023]	+0.2	+0.0007	[−0.0016, +0.0026]	+0.3
RM3	0.231	+0.0103	[+0.0077, +0.0131]	+4.5	+0.0076	[+0.0051, +0.0103]	+3.3	+0.0061	[+0.0036, +0.0077]	+2.6
Bo1	0.241	+0.0119	[+0.0081, +0.0157]	+4.9	+0.0043	[−0.0008, +0.0092]	+1.8	+0.0064	[+0.0027, +0.0100]	+2.6
Under 1M, base DPH										
CRVE S1	0.274	+0.0101	[+0.0075, +0.0127]	+3.7	+0.0052	[+0.0026, +0.0078]	+1.9	+0.0061	[+0.0042, +0.0085]	+2.2
CRVE S2	0.277	+0.0058	[+0.0031, +0.0085]	+2.1	+0.0014	[−0.0012, +0.0040]	+0.5	+0.0014	[−0.0004, +0.0025]	+0.5
RM3	0.262	+0.0137	[+0.0104, +0.0169]	+5.2	+0.0084	[+0.0057, +0.0112]	+3.2	+0.0086	[+0.0067, +0.0102]	+3.3
Bo1	0.252	+0.0102	[+0.0057, +0.0150]	+4.1	+0.0035	[−0.0004, +0.0073]	+1.4	+0.0027	[−0.0003, +0.0061]	+1.1
Over 1M, base BM25										
CRVE S1	0.367	+0.0041	[+0.0019, +0.0063]	+1.1	+0.0022	[−0.0002, +0.0046]	+0.6	+0.0023	[+0.0001, +0.0034]	+0.6
CRVE S2	0.367	+0.0031	[+0.0008, +0.0053]	+0.8	−0.0009	[−0.0030, +0.0012]	−0.2	−0.0001	[−0.0029, +0.0019]	0.0
RM3	0.346	+0.0170	[+0.0137, +0.0204]	+4.9	+0.0159	[+0.0126, +0.0191]	+4.6	+0.0158	[+0.0145, +0.0168]	+4.6
Bo1	0.350	+0.0058	[+0.0025, +0.0092]	+1.6	+0.0045	[+0.0014, +0.0077]	+1.3	+0.0024	[+0.0006, +0.0045]	+0.7
Over 1M, base DPH										
CRVE S1	0.406	+0.0065	[+0.0034, +0.0095]	+1.6	+0.0032	[0.0000, +0.0064]	+0.8	+0.0039	[+0.0019, +0.0056]	+1.0
CRVE S2	0.408	+0.0062	[+0.0031, +0.0094]	+1.5	+0.0042	[+0.0013, +0.0072]	+1.0	+0.0036	[+0.0013, +0.0053]	+0.9
RM3	0.397	+0.0137	[+0.0104, +0.0169]	+3.4	+0.0107	[+0.0075, +0.0139]	+2.7	+0.0118	[+0.0099, +0.0133]	+3.0
Bo1	0.386	+0.0085	[+0.0040, +0.0129]	+2.2	+0.0070	[+0.0030, +0.0109]	+1.8	+0.0054	[−0.0006, +0.0085]	+1.4

Table S4: Risk of tuned arms against our base retriever, in nDCG@10, on capped query sets (at most 500 queries per collection). Setting: default parameters, oracle (chosen on the test queries) or cross-validated (CV, seed-42 split). Below, above: collections more than 0.005 below or above the base. Worst: the collection with the lowest gain, with its change as a percentage of that collection's base level. Mean gain: macro mean, with its percentage of the group's base level (Rel.).

Method	Setting	< -0.005	$> +0.005$	Worst collection	Worst gain	Rel. (%)	Mean gain	Rel. (%)
Under 1M, base BM25: 20 collections, macro nDCG@10 0.242								
CRVE S1	default	1	12	Quora	−0.0122	−1.6	+0.0091	+3.8
CRVE S1	oracle	0	15	Quora	−0.0003	0.0	+0.0168	+6.9
CRVE S1	CV (seed 42)	2	11	TREC-COVID	−0.0235	−3.7	+0.0097	+4.0
CRVE S2	default	1	13	Quora	−0.0091	−1.2	+0.0091	+3.8
CRVE S2	oracle	0	14	Stack Overflow	−0.0010	−0.6	+0.0140	+5.8
CRVE S2	CV (seed 42)	1	12	Biology	−0.0072	−10.7	+0.0096	+4.0
RM3	default	16	3	Economics	−0.0451	−31.0	−0.0112	−4.6
RM3	oracle	11	5	Economics	−0.0306	−21.0	−0.0008	−0.3
RM3	CV (seed 42)	11	5	Economics	−0.0345	−23.7	−0.0035	−1.5
Bo1	default	7	11	ArguAna	−0.0707	−14.5	−0.0009	−0.4
Bo1	oracle	6	11	ArguAna	−0.0417	−8.5	+0.0110	+4.6
Bo1	CV (seed 42)	11	7	SciFact	−0.0472	−6.8	+0.0035	+1.4
RM3/Bo1 best of both	oracle	5	12	ArguAna	−0.0171	−3.5	+0.0165	+6.8
Under 1M, base DPH: 20 collections, macro nDCG@10 0.265								
CRVE S1	default	1	13	Quora	−0.0233	−5.4	+0.0082	+3.1
CRVE S1	oracle	0	15	TheoremQA-Theorems	0.0000	–[a]	+0.0183	+6.9
CRVE S1	CV (seed 42)	1	11	TREC-COVID	−0.0077	−1.2	+0.0134	+5.1
CRVE S2	default	1	14	Quora	−0.0159	−3.7	+0.0114	+4.3
CRVE S2	oracle	0	15	TheoremQA-Theorems	0.0000	–[a]	+0.0172	+6.5
CRVE S2	CV (seed 42)	0	11	Touché-2020	−0.0024	−0.6	+0.0128	+4.8
RM3	default	11	6	Quora	−0.0475	−11.0	−0.0035	−1.3
RM3	oracle	3	12	Quora	−0.0264	−6.1	+0.0102	+3.8
RM3	CV (seed 42)	8	8	Quora	−0.0264	−6.1	+0.0049	+1.9
Bo1	default	11	4	Quora	−0.0812	−18.7	−0.0129	−4.9
Bo1	oracle	10	8	Quora	−0.0666	−15.4	−0.0027	−1.0
Bo1	CV (seed 42)	13	5	Quora	−0.0668	−15.4	−0.0095	−3.6
RM3/Bo1 best of both	oracle	3	12	Quora	−0.0264	−6.1	+0.0114	+4.3
Over 1M, base BM25: 5 collections, macro nDCG@10 0.362								
CRVE S1	default	0	3	DBPedia-Entity	−0.0026	−0.8	+0.0050	+1.4
CRVE S1	oracle	0	3	DBPedia-Entity	+0.0028	+0.9	+0.0090	+2.5
CRVE S1	CV (seed 42)	0	3	HotpotQA	−0.0005	−0.1	+0.0071	+2.0
CRVE S2	default	0	2	HotpotQA	+0.0005	+0.1	+0.0044	+1.2
CRVE S2	oracle	0	3	DBPedia-Entity	+0.0042	+1.3	+0.0075	+2.1
CRVE S2	CV (seed 42)	0	2	DBPedia-Entity	−0.0003	−0.1	+0.0035	+1.0
RM3	default	3	1	HotpotQA	−0.0608	−10.4	−0.0165	−4.6
RM3	oracle	2	2	HotpotQA	−0.0287	−4.9	+0.0004	+0.1
RM3	CV (seed 42)	2	2	HotpotQA	−0.0304	−5.2	−0.0007	−0.2
Bo1	default	2	1	HotpotQA	−0.0498	−8.5	−0.0118	−3.3
Bo1	oracle	2	2	HotpotQA	−0.0436	−7.4	−0.0060	−1.7
Bo1	CV (seed 42)	2	2	HotpotQA	−0.0457	−7.8	−0.0072	−2.0
RM3/Bo1 best of both	oracle	2	2	HotpotQA	−0.0287	−4.9	+0.0017	+0.5
Over 1M, base DPH: 5 collections, macro nDCG@10 0.408								
CRVE S1	default	2	2	DBPedia-Entity	−0.0141	−4.5	−0.0020	−0.5
CRVE S1	oracle	0	2	DBPedia-Entity	−0.0011	−0.4	+0.0045	+1.1
CRVE S1	CV (seed 42)	0	1	NQ	−0.0025	−0.9	+0.0012	+0.3
CRVE S2	default	3	1	NQ	−0.0080	−2.7	−0.0001	0.0
CRVE S2	oracle	0	2	HotpotQA	+0.0002	0.0	+0.0061	+1.5
CRVE S2	CV (seed 42)	0	1	DBPedia-Entity	0.0000	0.0	+0.0042	+1.0
RM3	default	2	1	HotpotQA	−0.0562	−9.1	−0.0110	−2.7
RM3	oracle	1	2	HotpotQA	−0.0236	−3.8	+0.0027	+0.7
RM3	CV (seed 42)	1	2	HotpotQA	−0.0258	−4.2	−0.0003	−0.1
Bo1	default	4	1	HotpotQA	−0.0625	−10.1	−0.0227	−5.6
Bo1	oracle	3	1	HotpotQA	−0.0499	−8.1	−0.0142	−3.5
Bo1	CV (seed 42)	3	1	HotpotQA	−0.0499	−8.1	−0.0157	−3.9
RM3/Bo1 best of both	oracle	1	2	HotpotQA	−0.0236	−3.8	+0.0036	+0.9

[a] Undefined: the base nDCG@10 of this collection is zero.

S2 Cost of the Research Implementation

Table S5 gives the offline build cost per collection and Table S6 the query latency on the experiment machine. Our latency is that of a research implementation (Python, through PyTerrier) set against Terrier's native PRF implementations. Most of the overhead of CRVE over BM25 is implementation cost, in the per-call PyTerrier overhead of the lexical profile channel of Gate 1 and in the Python-side query encoding, rather than cost inherent to the method: the retrieval with the five added terms itself costs about 5 to 17 ms more than BM25 when the query is encoded on the CPU. Native implementations of Gate 1 and Gate 2 are expected to narrow the gap to PRF. LLM-Q2E takes 604 to 757 ms per query; the timed generations are fresh and differ from the cached generations of the effectiveness runs on 44% of the timed queries. On the five large collections the build figures are lower bounds, since the co-occurrence lists were computed only for the anchors of the evaluated queries.

Table S5: Offline build cost of CRVE per collection: build time, peak memory and extra disk. The figures use randomly drawn chunks (seed 42), not S2-selected chunks. Build: total time in minutes, of which encoding in seconds; Peak RSS: peak resident set size (RSS), the resident memory of the build; Extra disk: size of the CRVE files, also as a percentage of the Terrier index; Chunks: stored chunks.

Collection	Documents	Build (min)	Encoding (s)	Peak RSS (GiB)	Extra disk (MB)	Index (MB)	Disk / index (%)	Chunks
Under one million documents								
SciFact	5,183	0.8	7.2	2.378	61.1	25.5	239.4	18,942
NFCorpus	3,633	0.5	5.4	2.980	44.4	18.6	238.9	13,755
TREC-COVID	171,331	11.6	49.2	5.186	312.4	287.0	108.9	154,343
FiQA	57,600	3.5	26.0	5.041	159.2	84.0	189.7	82,847
Quora	522,931	3.0	29.8	4.594	153.4	212.6	72.1	158,730
ArguAna	8,674	0.8	8.1	2.482	64.2	25.6	250.9	20,130
SCIDOCS	25,657	2.5	20.6	3.520	137.8	69.0	199.7	58,443
Touché-2020	382,545	35.9	57.1	7.628	405.2	685.1	59.2	183,630
AoPS	188,002	8.8	39.6	4.778	173.2	165.7	104.5	125,773
Economics	50,220	1.9	16.6	5.298	114.2	168.8	67.7	54,709
LeetCode	413,932	19.2	57.2	5.888	313.6	577.6	54.3	180,836
Stack Overflow	107,081	9.3	27.4	5.248	154.3	285.1	54.1	87,527
Sustainable Living	60,792	2.0	18.3	5.311	118.2	193.5	61.1	60,160
Biology	57,359	1.7	16.1	6.206	113.5	149.1	76.1	53,026
Earth Science	121,249	3.7	31.4	6.730	182.0	330.7	55.0	100,419
Psychology	52,835	2.1	18.0	6.683	126.5	191.0	66.2	59,872
Robotics	61,961	1.6	16.2	6.650	88.9	74.3	119.7	54,080
Pony	7,894	0.2	3.6	6.504	12.1	8.3	145.6	7,693
TheoremQA-Questions	188,002	8.8	40.1	6.617	173.2	165.7	104.5	125,773
TheoremQA-Theorems	23,839	1.3	12.0	6.593	43.1	23.8	181.0	33,293
Over one million documents (lower bounds†)								
NQ	2,681,468	62.4†	59.8	2.476†	324.7	2304.9	14.1	193,789
DBPedia-Entity	4,635,922	72.6†	59.7	2.453†	302.5	4041.0	7.5	194,873
HotpotQA	5,233,329	78.3†	60.6	2.430†	335.3	4178.7	8.0	195,537
FEVER	5,416,568	132.9†	61.9	2.657†	355.5	5555.7	6.4	197,098
Climate-FEVER	5,416,593	133.6†	62.2	2.515†	349.3	5555.7	6.3	197,099

† Lower bounds: the co-occurrence lists are restricted to the anchors of the tested queries, and the peak memory excludes the PPMI step, which was not measured, because the full co-occurrence lists do not fit the RAM budget with the current code.

Table S6: Query latency of BM25, its expansion variants and CRVE on five collections (milliseconds per query, one query at a time; p90 denotes the 90th percentile). Implementation caveats: Section S2. CRVE is timed with randomly drawn chunks (seed 42); Gate 2 scores at most $L \cdot n$ stored vectors whichever score chose them, and the S2 sidecar was not timed. Number of queries timed in brackets after each collection.

Median (p90), ms	FiQA (100)	Quora (100)	Biology (53)	NQ (100)	HotpotQA (100)
Total time per query, with BM25 as base					
BM25	24.2 (26.2)	13.7 (16.1)	16.8 (19.1)	28.2 (38.4)	46.5 (81.5)
RM3	33.5 (35.5)	22.3 (26.3)	27.3 (31.5)	63.7 (88.1)	105.3 (185.1)
Bo1	32.3 (34.2)	22.3 (26.2)	26.5 (36.0)	55.3 (78.9)	94.7 (177.9)
BGE-QE[a]	33.2 (35.9)	21.1 (24.5)	27.7 (33.4)	48.7 (66.5)	81.7 (126.2)
LLM-Q2E[b]	603.5 (749.7)	681.7 (835.2)	757.0 (924.5)	660.3 (840.4)	666.1 (821.7)
CRVE, encoding on CPU	78.3 (85.3)	79.1 (86.9)	92.3 (105.2)	79.0 (98.8)	101.1 (140.4)
CRVE, encoding on GPU[c]	66.2 (73.2)	52.4 (62.4)	67.4 (76.5)	73.8 (90.9)	108.3 (230.1)
Stages of CRVE (median, ms)					
Gate 1, CPU	21.6	20.1	29.5	20.7	19.2
Query encoding, CPU	19.8	33.1	33.2	12.8	10.3
Gate 2, CPU	5.0	6.2	4.9	5.2	5.3
Retrieval, CPU	30.5	18.8	21.5	39.4	63.3
Gate 1, GPU[c]	21.7	19.9	28.0	20.9	19.1
Query encoding, GPU[c]	7.6	7.0	8.8	7.4	8.1
Gate 2, GPU[c]	5.0	5.5	5.0	5.5	5.1
Retrieval, GPU[c]	30.6	18.1	21.6	39.6	71.1

[a] BGE-QE encodes the query on the GPU, as in its pipeline, and the encoding is included. [b] qwen3.5:4b through Ollama, a fresh generation for every timed query; the model is reloaded every 20 queries. [c] From a separate timing session, not the session of the other rows; the CPU configuration measured in that session gives 78 to 103 ms; on HotpotQA the GPU row (108.3 ms) is slower than the CPU row (101.1 ms) because of variation in retrieval. The top five terms are the same as with the CPU configuration for every query.